CAS SciFinder

Task History

Initiating Search

November 6, 2025, 11:26 AM

Search:

CAS Lexicon Search:

Beckmann rearrangement - Preferred Concept

Search Tasks

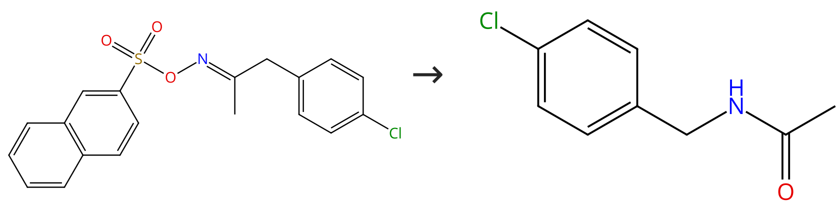
Task	Result Type	View
Returned Reference Results from CAS Lexicon (2,454)	References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,589) Filtered By: Yield: **90-100%, 80-89%, 70-79%** Document Type: **Journal** Language: **English**	Reactions	View Results

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (25)

31-452-CAS-604089

Steps: **1** Yield: **100%**

1.1 **Reagents:** *N,N*-Dimethylaniline
Solvents: Acetonitrile, Water; 25 °C

Experimental Protocols

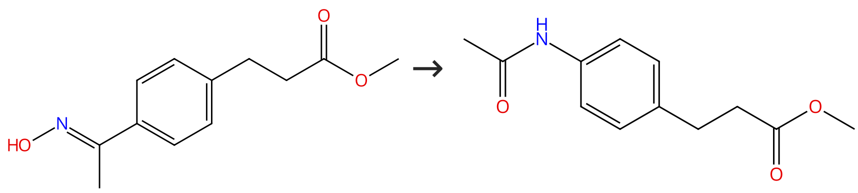
Dynamic Path Bifurcation in the Beckmann Reaction: Support from Kinetic Analyses

By: Yamamoto, Yutaro; et al

Journal of Organic Chemistry (2011), 76(11), 4652-4660.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (5)

31-452-CAS-18724100

Steps: **1** Yield: **100%**

1.1 **Catalysts:** 1,1,1,4,4,4-Hexafluoro-2,3-bis(trifluoromethyl)-2,3-butanediol, 1-Methyl 2-boronobenzoate
Solvents: Nitromethane, 1,1,1,3,3,3-Hexafluoro-2-propanol; 24 h, rt

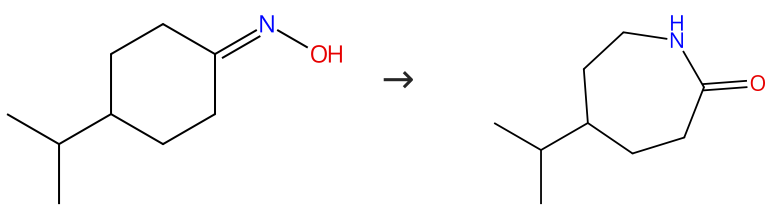
Scope and Mechanism of a True Organoca talytic Beckmann Rearrangement with a Boronic Acid/Pe rfluoropinacol System under Ambient Conditions

By: Mo, Xiaobin; et al

Journal of the American Chemical Society (2018), 140(15), 5264-5271.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (8)

Suppliers (30)

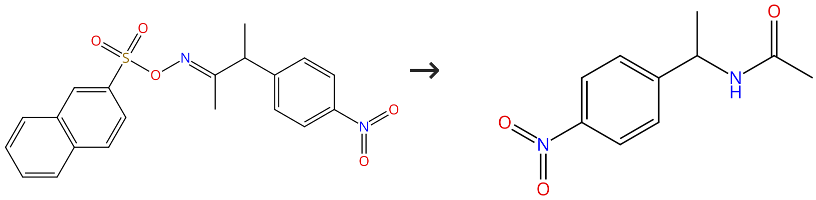
31-452-CAS-13267657	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Silica, Phosphorus oxychloride **Solvents:** Tetrahydrofuran; 5 min

Silica-supported dichlorophosphate catalyzed Beckmann rearrangement and dehydration of oximes under microwave irradiation

By: Li, Zheng; et al

Letters in Organic Chemistry (2008), 5(6), 495-501.

Scheme 4 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (29)

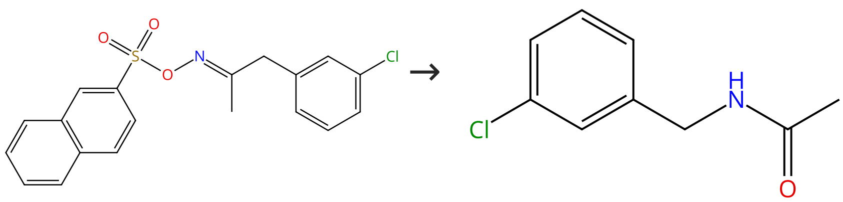
31-452-CAS-6185927	Steps: **1** Yield: **100%**
1.1	**Reagents:** *N*,*N*-Dimethylaniline **Solvents:** Acetonitrile, Water; 25 °C
Experimental Protocols	

Dynamic Path Bifurcation in the Beckmann Reaction: Support from Kinetic Analyses

By: Yamamoto, Yutaro; et al

Journal of Organic Chemistry (2011), 76(11), 4652-4660.

Scheme 5 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (13)

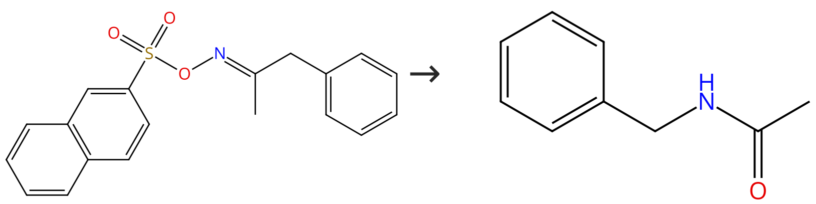
31-452-CAS-2743047	Steps: **1** Yield: **100%**
1.1	**Reagents:** *N*,*N*-Dimethylaniline **Solvents:** Acetonitrile, Water; 25 °C
Experimental Protocols	

Dynamic Path Bifurcation in the Beckmann Reaction: Support from Kinetic Analyses

By: Yamamoto, Yutaro; et al

Journal of Organic Chemistry (2011), 76(11), 4652-4660.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (2) Suppliers (78)

31-452-CAS-14003723 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N,N*-Dimethylaniline
Solvents: Acetonitrile, Water; 25 °C

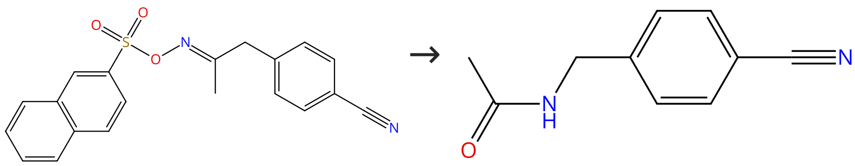
Experimental Protocols

Dynamic Path Bifurcation in the Beckmann Reaction: Support from Kinetic Analyses

By: Yamamoto, Yutaro; et al

Journal of Organic Chemistry (2011), 76(11), 4652-4660.

Scheme 7 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (3) Suppliers (9)

31-452-CAS-4491873 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N,N*-Dimethylaniline
Solvents: Acetonitrile, Water; 25 °C

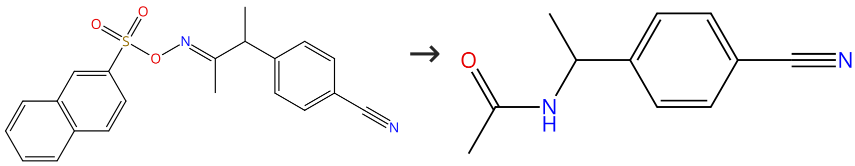
Experimental Protocols

Dynamic Path Bifurcation in the Beckmann Reaction: Support from Kinetic Analyses

By: Yamamoto, Yutaro; et al

Journal of Organic Chemistry (2011), 76(11), 4652-4660.

Scheme 8 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (8)

31-452-CAS-4335942 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N,N*-Dimethylaniline
Solvents: Acetonitrile, Water; 25 °C

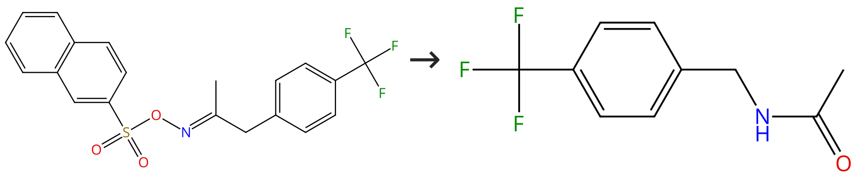
Experimental Protocols

Dynamic Path Bifurcation in the Beckmann Reaction: Support from Kinetic Analyses

By: Yamamoto, Yutaro; et al

Journal of Organic Chemistry (2011), 76(11), 4652-4660.

Scheme 9 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (11)

31-452-CAS-4927045 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N,N*-Dimethylaniline
Solvents: Acetonitrile, Water; 25 °C

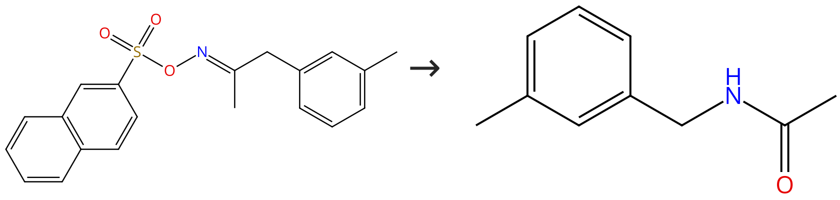
Experimental Protocols

Dynamic Path Bifurcation in the Beckmann Reaction: Support from Kinetic Analyses

By: Yamamoto, Yutaro; et al

Journal of Organic Chemistry (2011), 76(11), 4652-4660.

Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (10)

31-452-CAS-11873995 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N,N*-Dimethylaniline
Solvents: Acetonitrile, Water; 25 °C

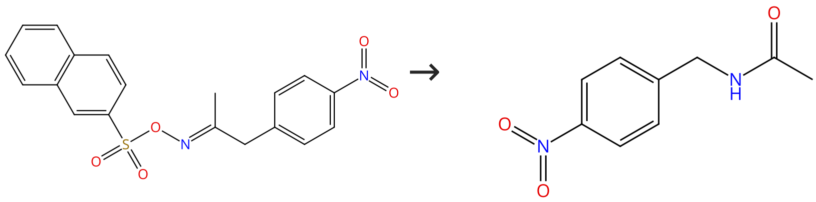
Experimental Protocols

Dynamic Path Bifurcation in the Beckmann Reaction: Support from Kinetic Analyses

By: Yamamoto, Yutaro; et al

Journal of Organic Chemistry (2011), 76(11), 4652-4660.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (37)

31-452-CAS-6625372 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N,N*-Dimethylaniline
Solvents: Acetonitrile, Water; 25 °C

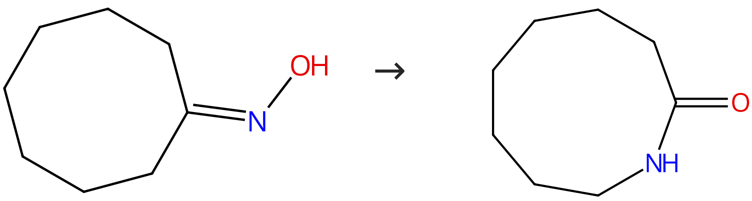
Experimental Protocols

Dynamic Path Bifurcation in the Beckmann Reaction: Support from Kinetic Analyses

By: Yamamoto, Yutaro; et al

Journal of Organic Chemistry (2011), 76(11), 4652-4660.

Scheme 12 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (41) Suppliers (69)

31-452-CAS-5204147 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Aluminum chloride

Experimental Protocols

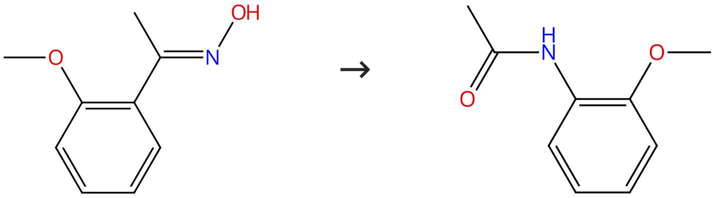
A facile Beckmann rearrangement of oximes with $AlCl_3$ in the solid state

By: Ghiaci, M.; et al

Synthetic Communications (1998), 28(12), 2275-2280.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (17)

Suppliers (60)

31-452-CAS-1446130

Steps: **1** Yield: **100%**

1.1 **Catalysts:** Silica (chlorinated, mesoporous)
Solvents: Toluene; 2 h, reflux

Experimental Protocols

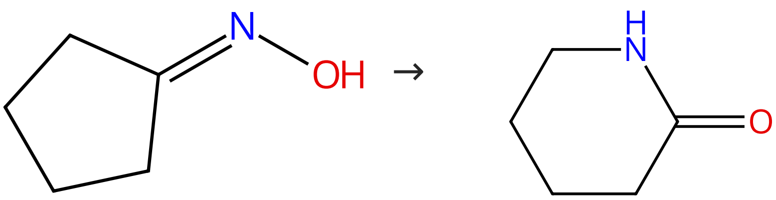
Novel periodic mesoporous silica chlorides (P MSCl) with 2D P6mm hexagonal structures: efficient catalysts for the Beckmann rearrangement

By: Karimi, Babak; et al

Synlett (2010), (13), 2019-2023.

Scheme 14 (2 Reactions)

Steps: **1** Yield: **100%**



Suppliers (65)

Suppliers (90)

31-452-CAS-8841769

Steps: **1** Yield: **100%**

1.1 **Catalysts:** Benzenesulfonic acid, 4-ethenyl-, sodium salt (1:1), polymer with diethenylbenzene (neutralized)
Solvents: Benzonitrile; 1 h, 130 °C

Experimental Protocols

In situ functionalized sulfonic copolymer toward recyclable heterogeneous catalyst for efficient Beckmann rearran gement of cyclohexanone oxime

By: Li, Difan; et al

Applied Catalysis, A: General (2016), 510, 125-133.

31-452-CAS-967743

Steps: **1** Yield: **100%**

1.1 **Catalysts:** Aluminum chloride

Experimental Protocols

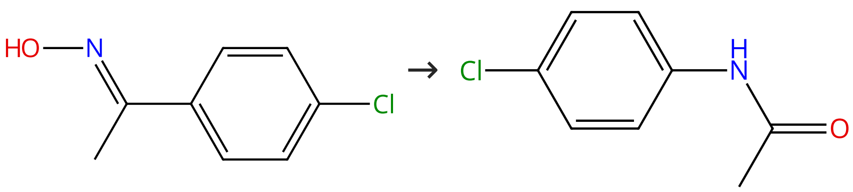
A facile Beckmann rearrangement of oximes with $AlCl_3$ in the solid state

By: Ghiaci, M.; et al

Synthetic Communications (1998), 28(12), 2275-2280.

Scheme 15 (4 Reactions)

Steps: **1** Yield: **100%**



Suppliers (34)

Suppliers (93)

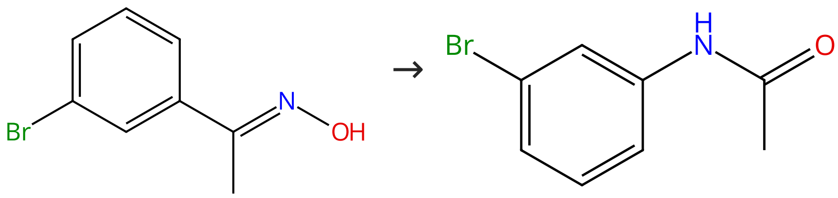
31-452-CAS-2163773	Steps: **1** Yield: **100%**	In situ functionalized sulfonic copolymer toward recyclable heterogeneous catalyst for efficient Beckmann rearran gement of cyclohexanone oxime
1.1	**Catalysts:** Benzenesulfonic acid, 4-ethenyl-, sodium salt (1:1), polymer with diethenylbenzene (neutralized) **Solvents:** Benzonitrile; 1 h, 130 °C	By: Li, Difan; et al
Experimental Protocols		Applied Catalysis, A: General (2016), 510, 125-133.

31-452-CAS-14985730	Steps: **1** Yield: **100%**	Imidazolium and pyridinium salts-solvents influencing the rate and direction of the Fries, Beckmann, and Claisen rearrangements
1.1	**Catalysts:** Tin tetrachloride **Solvents:** 1-Heptylpyridinium bromide; rt; 30 min, 120 °C	By: Katkevica, S.; et al
1.2	3 h, 120 °C	Chemistry of Heterocyclic Compounds (New York, NY, United States) (2010), 46(2), 158-169.
Experimental Protocols		

31-452-CAS-12630810	Steps: **1** Yield: **100%**	Lewis acid-catalyzed Beckmann rearrangement of ketoximes in ionic liquids
1.1	**Catalysts:** Aluminum chloride **Solvents:** 1*H*-Imidazolium, 3-butyl-1,2-dimethyl-, bromide (1:1) ; 30 min, 80 °C	By: Zicmanis, Andris; et al
		Catalysis Communications (2009), 10(5), 614-619.

31-452-CAS-14781791	Steps: **1** Yield: **100%**	Lewis acid-catalyzed Beckmann rearrangement of ketoximes in ionic liquids
1.1	**Catalysts:** Boron trifluoride **Solvents:** Pyridinium, 1-heptyl-, hexafluorophosphate(1-) (1:1); 3 h, 120 °C	By: Zicmanis, Andris; et al
		Catalysis Communications (2009), 10(5), 614-619.

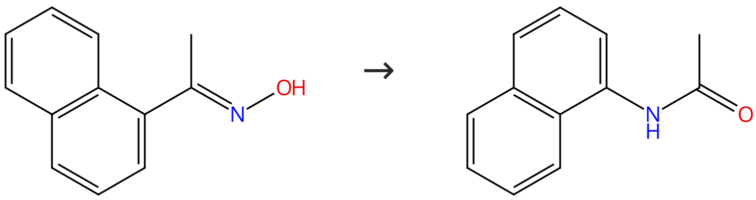
Scheme 16 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (10) Suppliers (59)

31-452-CAS-151787	Steps: **1** Yield: **100%**	Silica-supported dichlorophosphate catalyzed Beckmann rearrangement and dehydration of oximes under microwave irradiation
1.1	**Catalysts:** Silica, Phosphorus oxychloride **Solvents:** Tetrahydrofuran; 5 min	By: Li, Zheng; et al
		Letters in Organic Chemistry (2008), 5(6), 495-501.

Scheme 17 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (38) Suppliers (61)

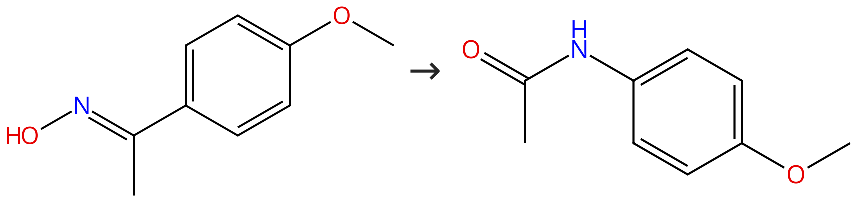
31-452-CAS-1384401	Steps: **1** Yield: **100%**
1.1 **Catalysts:** Tin tetrachloride **Solvents:** Tetrabutylammonium bromide; 180 min, 80 °C	

Lewis acid-catalyzed Beckmann rearrangement of ketoximes in ionic liquids

By: Zicmanis, Andris; et al

Catalysis Communications (2009), 10(5), 614-619.

Scheme 18 (3 Reactions) Steps: **1** Yield: **100%**



Suppliers (20) Suppliers (82)

31-452-CAS-13437207	Steps: **1** Yield: **100%**
1.1 **Catalysts:** Benzenesulfonic acid, 4-ethenyl-, sodium salt (1:1), polymer with diethenylbenzene (neutralized) **Solvents:** Benzonitrile; 1 h, 130 °C	
Experimental Protocols	

In situ functionalized sulfonic copolymer toward recyclable heterogeneous catalyst for efficient Beckmann rearran gement of cyclohexanone oxime

By: Li, Difan; et al

Applied Catalysis, A: General (2016), 510, 125-133.

31-452-CAS-12856626	Steps: **1** Yield: **100%**
1.1 **Catalysts:** Tin tetrachloride **Solvents:** 1-Heptylpyridinium bromide; rt; 30 min, 120 °C	
1.2 3 h, 120 °C	
Experimental Protocols	

Imidazolium and pyridinium salts-solvents influencing the rate and direction of the Fries, Beckmann, and Claisen rearrangements

By: Katkevica, S.; et al

Chemistry of Heterocyclic Compounds (New York, NY, United States) (2010), 46(2), 158-169.

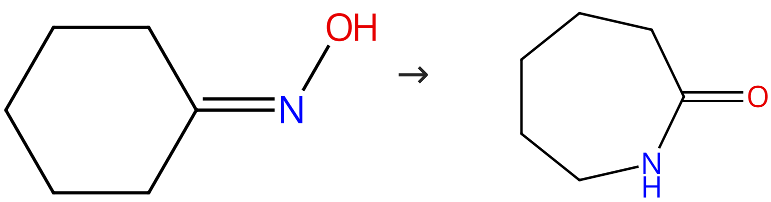
31-452-CAS-10518939	Steps: **1** Yield: **100%**
1.1 **Catalysts:** Aluminum chloride **Solvents:** 1-Butyl-3-methylimidazolium bromide; 30 min, 80 °C	

Lewis acid-catalyzed Beckmann rearrangement of ketoximes in ionic liquids

By: Zicmanis, Andris; et al

Catalysis Communications (2009), 10(5), 614-619.

Scheme 19 (24 Reactions) Steps: **1** Yield: **100%**



Suppliers (60) Suppliers (111)

31-614-CAS-36298771	Steps: **1** Yield: **100%**
1.1 **Catalysts:** Bis(hexamethylene)triamine **Solvents:** Octane; 20 min, 100 °C	
Experimental Protocols	

Beckmann Rearrangement with Improved Atom Economy, Catalyzed by Inexpensive, Reusable, Bronsted Acidic Ionic Liquid

By: Brzeczek-Szafran, Alina; et al

ACS Sustainable Chemistry & Engineering (2022), 10(41), 13568-13575.

31-614-CAS-38090788 Steps: **1** Yield: **100%** 1.1 **Catalysts:** 1,6-Hexanediaminium, *N*1-hexyl-*N*1,*N*1,*N*6,*N*6-tetramethyl-*N*6-octadecyl-, bromide (1:2) **Solvents:** Methanol; 32 h, 350 °C Experimental Protocols	**Controllable Synthesis and Structure-Performance Relationship of Silicalite-1 Nanosheets in Vapor Phase Beckmann Rearrangement of Cyclohexanone Oxime** By: Ge, Chao; et al Catalysis Letters (2021), 151(5), 1488-1498.
31-452-CAS-19873725 Steps: **1** Yield: **100%** 1.1 **Catalysts:** Pyrrolidine, 1-methyl-, sulfate (1:2); 15 min, 110 °C	**Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*** By: Matuszek, Karolina; et al Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.
31-452-CAS-19873727 Steps: **1** Yield: **100%** 1.1 **Catalysts:** 1*H*-Imidazole, 1-methyl-, sulfate (1:2); 15 min, 110 °C	**Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*** By: Matuszek, Karolina; et al Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.
31-452-CAS-19873722 Steps: **1** Yield: **100%** 1.1 **Catalysts:** Sulfuric acid, compd. with *N*,*N*-diethylethanamine (2:1); 15 min, 110 °C	**Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*** By: Matuszek, Karolina; et al Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.
31-452-CAS-19873723 Steps: **1** Yield: **100%** 1.1 **Catalysts:** Pyrrolidine, 1-methyl-, sulfate (1:1); 15 min, 110 °C	**Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*** By: Matuszek, Karolina; et al Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.
31-452-CAS-19873728 Steps: **1** Yield: **100%** 1.1 **Catalysts:** Sulfuric acid, compd. with *N*,*N*-diethylethanamine (3:1); 15 min, 110 °C	**Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*** By: Matuszek, Karolina; et al Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.
31-452-CAS-19873726 Steps: **1** Yield: **100%** 1.1 **Catalysts:** 2275643-63-3; 15 min, 110 °C	**Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*** By: Matuszek, Karolina; et al Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.
31-452-CAS-19873720 Steps: **1** Yield: **100%** 1.1 **Catalysts:** 1*H*-Imidazole, 1-methyl-, sulfate (1:3); 15 min, 110 °C	**Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*** By: Matuszek, Karolina; et al Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.
31-452-CAS-19873724 Steps: **1** Yield: **100%** 1.1 **Catalysts:** 1-Methylimidazolium hydrogen sulfate; 15 min, 110 °C	**Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*** By: Matuszek, Karolina; et al Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.

31-452-CAS-19873721 Steps: **1** Yield: **100%**

1.1 **Catalysts:** 2275643-62-2; 15 min, 110 °C

Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*

By: Matuszek, Karolina; et al

Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.

31-452-CAS-19873729 Steps: **1** Yield: **100%**

1.1 **Catalysts:** 2275643-66-6; 15 min, 110 °C

Protic Ionic Liquids Based on Oligomeric Anions $[(HSO_4)(H_2SO_4)_x]^-$ (x=0, 1, or 2) for a Clean ε- Caprolactam Synthesis*

By: Matuszek, Karolina; et al

Australian Journal of Chemistry (2019), 72(1 & 2), 130-138.

31-452-CAS-16953516 Steps: **1** Yield: **100%**

1.1 **Solvents:** Benzonitrile; 13.4 min, 0.95 atm, 130 °C

Experimental Protocols

Understanding the Role of Molecular Diffusion and Catalytic Selectivity in Liquid-Phase Beckmann Rearrangement

By: Potter, Matthew E.; et al

ACS Catalysis (2017), 7(4), 2926-2934.

31-452-CAS-16195491 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Indium
Solvents: Benzene; 623 K

Experimental Protocols

Highly stable In-SBA-15 catalyst for vapor phase Beckmann rearrangement reaction

By: Kumar, Rawesh; et al

Microporous and Mesoporous Materials (2016), 234, 293-302.

31-452-CAS-9119442 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Benzenesulfonic acid, 4-ethenyl-, sodium salt (1:1), polymer with diethenylbenzene (neutralized)
Solvents: Benzonitrile; 1 h, 130 °C

Experimental Protocols

In situ functionalized sulfonic copolymer toward recyclable heterogeneous catalyst for efficient Beckmann rearrangement of cyclohexanone oxime

By: Li, Difan; et al

Applied Catalysis, A: General (2016), 510, 125-133.

31-452-CAS-6597082 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Bismuth, Silica; 623 K

Experimental Protocols

Bismuth supported SBA-15 catalyst for vapour phase Beckmann rearrangement reaction of cyclohexanone oxime to ε-caprolactam

By: Kumar, Rawesh; et al

Applied Catalysis, A: General (2015), 497, 51-57.

31-452-CAS-2056805 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Trifluoroacetic acid
Solvents: Acetonitrile; 48 s, 100 °C

1.2 **Solvents:** Acetonitrile; cooled

Experimental Protocols

Organocatalyzed Beckmann Rearrangement of Cyclohexanone Oxime in a Microchemical System

By: Zhang, Jisong; et al

Organic Process Research & Development (2015), 19(2), 352-356.

31-452-CAS-14831755 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Hexadecyltrimethylammonium bromide, Tetraethoxysilane, Sulfuric acid, Ammonium sulfate, Zirconium *n*-propoxide
Solvents: Acetonitrile; 3 h, 150 °C

Zr mesoporous molecular sieves as novel solid acid catalysts in synthesizing nitrile and caprolactam

By: Nedumaran, D.; et al

Journal of Nanoscience and Nanotechnology (2014), 14(4), 2799-2809.

31-452-CAS-10467091 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Niobium, Silica
Solvents: Benzene; 60 min, 350 °C

Experimental Protocols

Niobium doped hexagonal mesoporous silica (HMS-X) catalyst for vapor phase Beckmann rearrangement reaction

By: Mandal, Sandip; et al

RSC Advances (2014), 4(2), 845-854.

31-452-CAS-3696755 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Cyanuric chloride, 1*H*-Imidazolium, 3-[2-(formylmethylamino)ethyl]-1-methyl-, salt with 1,1,1-trifluoro-*N*-[(trifluoromethyl)sulfonyl]methanesulfonamide (1:1)
Solvents: Acetonitrile; 25 °C

1.2 **Solvents:** Acetonitrile; 2 h, 25 °C

Preparation of ionic liquid-based Vilsmeier reagent from novel multi-purpose dimethylformamide-like ionic liquid and its application

By: Hullio, Ahmed Ali; et al

Chinese Journal of Chemistry (2012), 30(7), 1647-1657.

31-452-CAS-9462933 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Fuming sulfuric acid
Solvents: Octane; < 40 s, 70 °C

1.2 **Reagents:** Water

Beckmann rearrangement in a microstructured chemical system for the preparation of ε-caprolactam

By: Zhang, J. S.; et al

AIChE Journal (2012), 58(3), 925-931.

31-452-CAS-2761022 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sulfuric acid
Catalysts: Water
Solvents: Water; 293 K

Applications of environmentally benign supercritical water to organic syntheses

By: Sato, Masahiro; et al

Analytical Sciences (2006), 22(11), 1409-1416.

31-452-CAS-10489077 Steps: **1** Yield: **100%**

1.1 **Reagents:** Phosphorus pentoxide, Hydroxyamine hydrochloride, Silica; 3 min

Experimental Protocols

An easy method for the generation of amides from ketones by a Beckmann type rearrangement mediated by microwave

By: Eshghi, Hossein; et al

Synthetic Communications (2003), 33(17), 2971-2978.

31-452-CAS-9362560 Steps: **1** Yield: **100%**

1.1 **Reagents:** Formamide, *N*,*N*-dimethyl-, compd. with 2,4,6-trichloro-1,3,5-triazine (1:1)
Solvents: Dimethylformamide

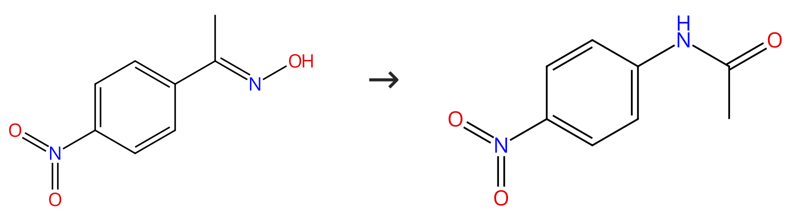
Experimental Protocols

Beckmann Rearrangement of Oximes under Very Mild Conditions

By: De Luca, Lidia; et al

Journal of Organic Chemistry (2002), 67(17), 6272-6274.

Scheme 20 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (25) Suppliers (75)

31-452-CAS-4587339 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Benzenesulfonic acid, 4-ethenyl-, sodium salt (1:1), polymer with diethenylbenzene (neutralized)
Solvents: Benzonitrile; 1 h, 130 °C

Experimental Protocols

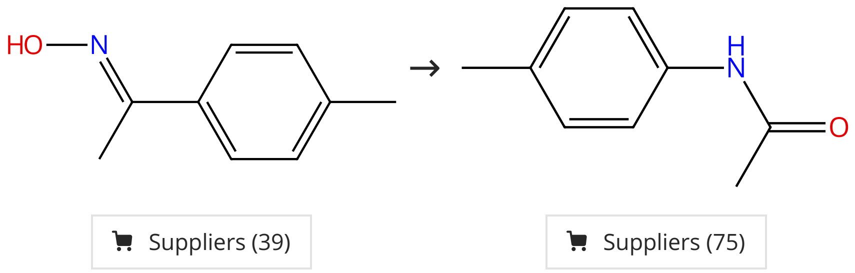
In situ functionalized sulfonic copolymer toward recyclable heterogeneous catalyst for efficient Beckmann rearrangement of cyclohexanone oxime

By: Li, Difan; et al

Applied Catalysis, A: General (2016), 510, 125-133.

Scheme 21 (1 Reaction)

Steps: **1** Yield: **100%**



31-452-CAS-11398123

Steps: **1** Yield: **100%**

1.1 **Catalysts:** Silica, Phosphorus oxychloride
Solvents: Tetrahydrofuran; 5 min

Silica-supported dichlorophosphate catalyzed Beckmann rearrangement and dehydration of oximes under microwave irradiation

By: Li, Zheng; et al

Letters in Organic Chemistry (2008), 5(6), 495-501.